



Review

Energy poverty in developing countries: A review of the concept and its measurements

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ARTICLE INFO

Keywords:

Energy poverty
Fuel poverty
Energy poverty indicators
Developing countries

ABSTRACT

This paper presents a meta-narrative review of the concept and measurements of energy poverty in developing countries. The literature survey was conducted using the Search, Appraisal, Synthesis, and Analysis (SALSA) and snowballing methods. Three academic databases, including Science Direct, Google Scholar, and Scopus, were searched for studies published in English from 1970 to 2021 on the topic of energy poverty in developing countries. The review found that many definitions are provided to explain the concept of energy poverty due to the differences between countries in terms of climate and season, household preferences, accommodating home standards, and social and cultural beliefs. The lack of generally accepted definition of energy poverty has led to confusion between the concepts of energy poverty and fuel poverty. Both concepts apply to the problem of energy consumption in residential sector, but they have similitudes and differences. Three categories of indicators are used for communicating energy poverty issues at the local, national and global levels: single indicators, dashboard indicators, and composite indices. Each approach has potential advantages and limitations. The composite indices are more generally applied because of their capacity to aggregate various dimensions of energy poverty and facilitate benchmarking between countries. However, the scarcity of data prevents the widespread use of the measurements of energy poverty. Consequently, it contributes to the delay in assessing the implementation of energy policy to achieve the sustainable development goal of eradicating energy poverty in developing countries.

1. Introduction

Energy poverty in developing countries is an especially complex issue that has been receiving important attention from researchers and international organizations, including the United Nations (UN), International Energy Agency (IEA), International Atomic Energy Agency (IAEA), and World Bank [1]. As energy is central to economic development, ending energy poverty is an inherently political process, which is at the heart of the 2030 agenda for sustainable development [2,3]. All countries, in particular developing countries, have set up diverse actions aiming at providing universal access to modern energy services, reducing global energy intensity, and increasing global renewable energy use to primary energy supply [4]. Nussbaumer et al. [5] affirm that if current actions continue to fall short both in terms of scale and pace, more people will lack access to modern energy services in 2030.

In the framework of assessing the implementation of the actions for eradicating energy poverty in developing countries, researchers and policymakers have provided various definitions of the concept of energy

poverty in the literature. Energy poverty is a highly controversial concept [6,7]. In fact, various publications confuse the concepts of energy poverty and fuel poverty, which apply to the problem of energy consumption in the residential sector [8]. Researchers and experts in the field of energy poverty in developing countries develop several indicators to monitor and report progress towards achieving sustainable development goals [9].

The present paper reviews the growing body of literature on energy poverty in developing countries. The focus on developing countries is essentially based on two reasons. First, energy poverty is an issue for both developed and developing countries, but within different contexts [10,11]. Energy poverty in developed countries is related to the issues of affordability and energy efficiency [12]. In developed countries, there is a greater infrastructure coverage and an equitable access to modern energy sources [13,14,15]. However, energy poverty in developing countries refers to a lack of access to modern energy sources, which are needed to provide modern energy services [8,15,16]. It can be explained by the lack of infrastructure, well-functioning energy services markets,

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and sufficient income to afford modern energy sources [14,17]. Energy poverty in developing countries affects the entire territorial units. The rural and remote areas remain particularly disadvantaged, with no or little access to modern energy sources.

The second reason is that knowledge production within the field of energy poverty in developing countries is increasing exponentially. The growing number of publications makes it hard to understand the conceptualizations and measurements of energy poverty, to know where the state-of-the art is and to be at the forefront of research [18]. Only a handful of studies have attempted to review the measurements of energy poverty in developing countries [19,20,21]. These studies focus on the indices for multidimensional energy poverty assessment at the micro-level. In fact, several indicators (e.g., unidimensional and target-based indicators) are proposed to assess energy poverty in developing countries at both the micro and macro-levels, but they are not taken into consideration. The existing studies do not track the development of measurements of energy poverty over time and thereby fail to present the state-of-the art of research.

The paper is intended to contribute to a better understanding of the lack of a generally accepted definition of energy poverty in developing countries. Furthermore, it supplements the existing studies by extending the literature on the measurements of energy poverty, and clarifying the current state of knowledge. It provides a meta-narrative review of the existing literature on the concept and measurements of energy poverty in developing countries. It gives a comprehensive overview of the conceptual background and tracks development of the methodologies over time [18]. Accordingly, the paper provides an entrée for researchers who are unfamiliar with the field of energy poverty in developing countries and allows the experts in this field to gain knowledge about the future directions for needed research.

This paper is organized as follows. The next section explains the methods behind the literature survey. Section 3 discusses the concept of energy poverty in developing countries, presents a synthesis of the measurements of energy poverty in developing countries and provides the state-of-the art of research. Finally, Section 4 provides some concluding remarks.

2. Methods

The meta-narrative review was applied using the Search, Appraisal, Synthesis, and Analysis (SALSA) and snowballing methods in order to explain historically how research traditions in the field of energy poverty in developing countries have unfolded over time and the methodologies used to assess energy poverty [22] (Fig. 1). The SALSA framework is a simple approach, which has logical sequence of steps for identifying all relevant publications falling within the scope of the research [23]. The snowballing technique identifies other relevant publications that are cited in the articles obtained using SALSA framework [20,21,23]. The combination of both methods minimizes the bias by conducting an exhaustive literature search [24]. In addition, this framework establishes a literature survey that is transparent and replicable [23,24].

2.1. Search

The following search strings were first entered in Science Direct, Google Scholar, and Scopus search engines (Fig. 2): “energy poverty” AND “developing countries”; “measuring energy poverty” AND “developing countries”. The literature search were conducted from 1970 to 2021 in the English language. Many publications gathered from Science Direct and Google Scholar databases using the search string “energy poverty” AND “developing countries” proved irrelevant to the research scope. Following Malinauskaite et al. [23], we considered different words that are often used as though they were synonyms of the search terms to define other search strings. The terms “energy access” and “fuel poverty” were used as synonyms for “energy poverty”. The term “developing countries” was associated with “developing world”, “global south”, “developing economies”, “Africa”, “Asia”, and “Latin America”. Two other search strings were used to identify potential relevant publications: “energy poverty” AND (“energy access” OR “fuel poverty”) AND “developing countries” AND (“developing world” OR “global south” OR “developing economies”); “measuring energy poverty” AND (“energy access” OR “fuel poverty”) AND “developing countries” AND (“Africa” OR “Asia” OR “Latin America”). The inquiry yielded 1177 articles from Science Direct, Google Scholar, and Scopus databases. After removing duplicates, seven hundred and twenty eight articles were taken to the appraisal step.

2.2. Appraisal and snowballing technique

The appraisal step consisted of screening a pool of eligible articles, using the snowball sampling, and selecting the relevant publications. The screening involved the application of inclusion-exclusion criteria by reading the title, abstract, and keywords of the publications gathered through the search strategies. The result was that 584 publications were excluded during the screening process. Thereafter, we assessed the full-text of all eligible publications. The snowball sampling was also applied to identify other potential relevant publications that are cited in the eligible publications. It yielded 74 publications. Overall, one hundred and seventy publications were deemed to fall within the scope of the research.

2.3. Synthesis

The synthesis required the categorization and description of all selected publications. There are different types of publications: 149 peer reviewed journal articles, 11 reports from inter-governmental and non-governmental organizations, and 10 publications from universities and governmental institutions. Furthermore, the publications are grouped under four themes such as energy poverty, fuel poverty, energy poverty and fuel poverty, and measuring energy poverty. Forty-five publications on the theme of energy poverty discuss the energy access issues in developing countries. Ten publications on the theme of fuel poverty deal with the issues of thermal comfort in the residential sector. The theme of energy poverty and fuel poverty encompasses three publications that

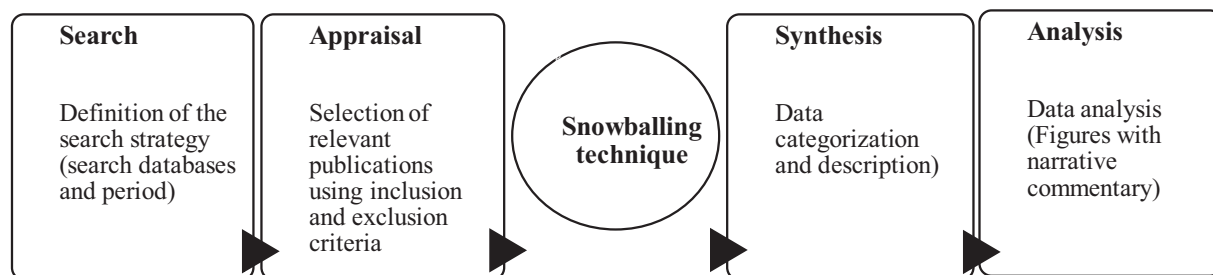


Fig. 1. Framework used for the meta-narrative review.
Source: Adapted from Malinauskaite et al. [23].

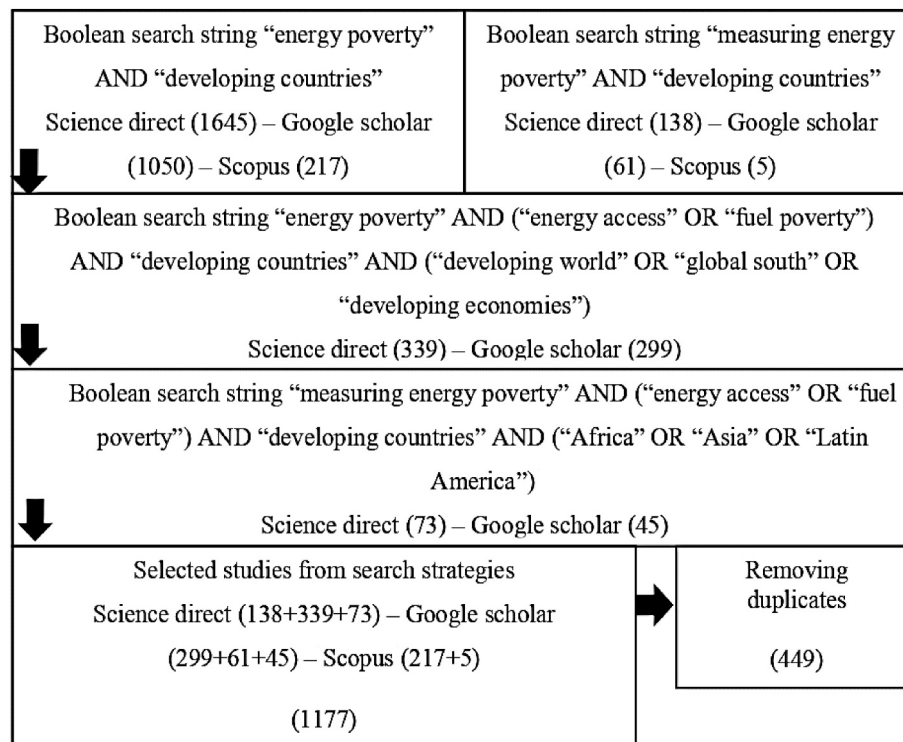


Fig. 2. Identification of potential relevant publications.
Source: Done by authors.

deal with the conceptual confusion in the literature. As regards the theme of measuring energy poverty, one hundred and twelve publications examine the methodologies already applied in developing countries to assess energy poverty.

2.4. Analysis

The analysis refers mainly to the following themes: energy poverty, energy poverty and fuel poverty, and measuring energy poverty. Overall, one hundred and fifty two reviewed publications were included in this analysis. Fig. 3 depicts the number per year of the reviewed publications. Few publications were done between 1978 and 2009. The start of the 2010s saw an increase in publications on energy poverty in developing countries. This increase is more pronounced since 2019. Approximately, 11%, 8%, and 24% of the reviewed studies were published in 2019, 2020, and 2021 respectively, resulting in 43% of the reviewed studies. Emphasis is also given to the peer-reviewed articles in order to identify the top journals contributing to the field of energy poverty in developing countries (Fig. 4). These journals publish new

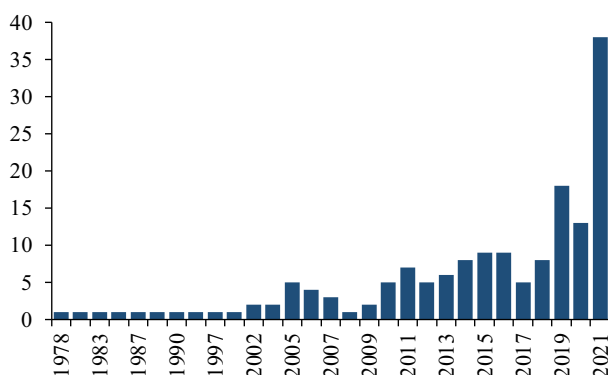


Fig. 3. Number per year of the reviewed publications.

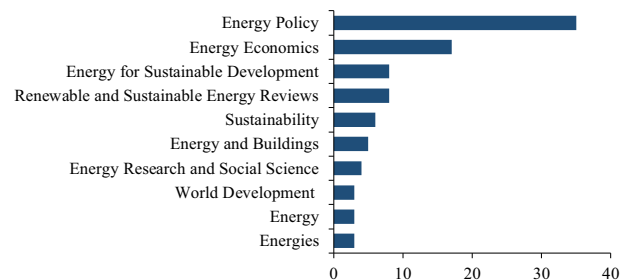


Fig. 4. Number per journal of the reviewed articles.
Source: Done by authors.

thinking on energy policy in developing countries and methodologies. Most articles are published in Energy Policy and Energy Economics journals.

The publications, in line with the theme of measuring energy poverty, examine energy poverty at both micro and macro-levels. The literature survey covers 91 developing countries from Africa, Asia, and Latin America and the Caribbean (Table 1). Twenty-eight methodologies are used to assess energy poverty. They are grouped in three broad categories: single indicators, dashboard indicators, and composite indices. Fig. 5 shows the spatial distribution of publications on measuring energy poverty. Several studies are conducted to assess energy poverty at both micro and macro-levels in Bangladesh, Cambodia, China, Ethiopia, Ghana, Kenya, India, Indonesia, Mexico, Nepal, Nigeria, Pakistan, Peru, Philippines, Senegal, South Africa, and Vietnam. There are many publications on assessing energy poverty in countries like Argentina, Benin, Brazil, Cameroon, Egypt, Malaysia, Morocco, Mozambique, Sri Lanka, Thailand, Tunisia, Uruguay, Zambia, and Zimbabwe, but a very few of them deal with the issues of energy poverty at the micro-level. The reviewed studies are not carried out for the purpose of assessing energy poverty at the micro-level in countries like

Table 1
Regions and countries covered by the literature survey.

Regions	Countries
Africa	Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Congo, Côte d'Ivoire, Democratic Republic of Congo, Egypt, Eritrea, Eswatini, Ethiopia, Gabon, Ghana, Guinea, Kenya, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Morocco, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, Togo, Tunisia, Uganda, Zambia, and Zimbabwe
Asia	Afghanistan, Azerbaijan, Bangladesh, Bhutan, Cambodia, China, India, Indonesia, Iran, Iraq, Jordan, Kazakhstan, Kyrgyz Republic, Laos, Lebanon, Malaysia, Maldives, Myanmar, Nepal, Philippines, Sri Lanka, Syria, Tajikistan, Thailand, Timor Leste, Uzbekistan, Vietnam, and Yemen
Latin America and the Caribbean	Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela

Algeria, Côte d'Ivoire, El Salvador, Eritrea, Gabon, Iran, Jamaica, Kazakhstan, Kyrgyz Republic, Lebanon, Libya, Panama, Paraguay, Sudan, Syria, Uzbekistan, Venezuela, and Yemen.

Finally, we emphasize that fifty-five out of one hundred and twelve publications in line with the theme of measuring energy poverty were not included in the results and discussion sections of this review because they provide the replication of the measurements of energy poverty. Hence, the paper reviews 115 publications in line with the four themes.

3. Results and discussion

3.1. Concept of energy poverty in developing countries

The origin of the concept of energy poverty dates from 1970. This concept has captured the attention of an increasing number of researchers and policymakers in the early 2010s [25]. A starting point for an enhanced understanding of energy poverty in developing countries is to raise readers' awareness regarding the difficulties in providing a single definition of the concept of energy access. The lack of consensus on how to interpret basic energy services has given rise to the controversy surrounding the definition of the concept of energy access, thereby leading to the lack of a generally accepted definition of energy poverty [19].

3.1.1. Concept of energy access

For a long time, energy access was defined as household electricity access [26]. There is no universally accepted definition for electricity access [27]. Electricity access can refer to physical availability or access to the grid [28]. In fact, developing countries were concerned about enhancing universal access to electricity. There has been a delay in implementing policies aimed at ensuring access to other energy sources needed to meet basic energy services such as cooking and water heating. Thereafter, the concept of energy access was defined in terms of physical availability of energy, namely traditional (e.g., firewood and agricultural residue) and modern (e.g., electricity and LPG) energy sources [28]. Modern energy sources are more efficient and environmentally sustainable than traditional ones. Accordingly, Ranjit and O'Sullivan [29] define the concept of energy access as the physical availability of modern energy sources. These definitions capture only the availability dimension of energy access. The concept of energy access takes complex

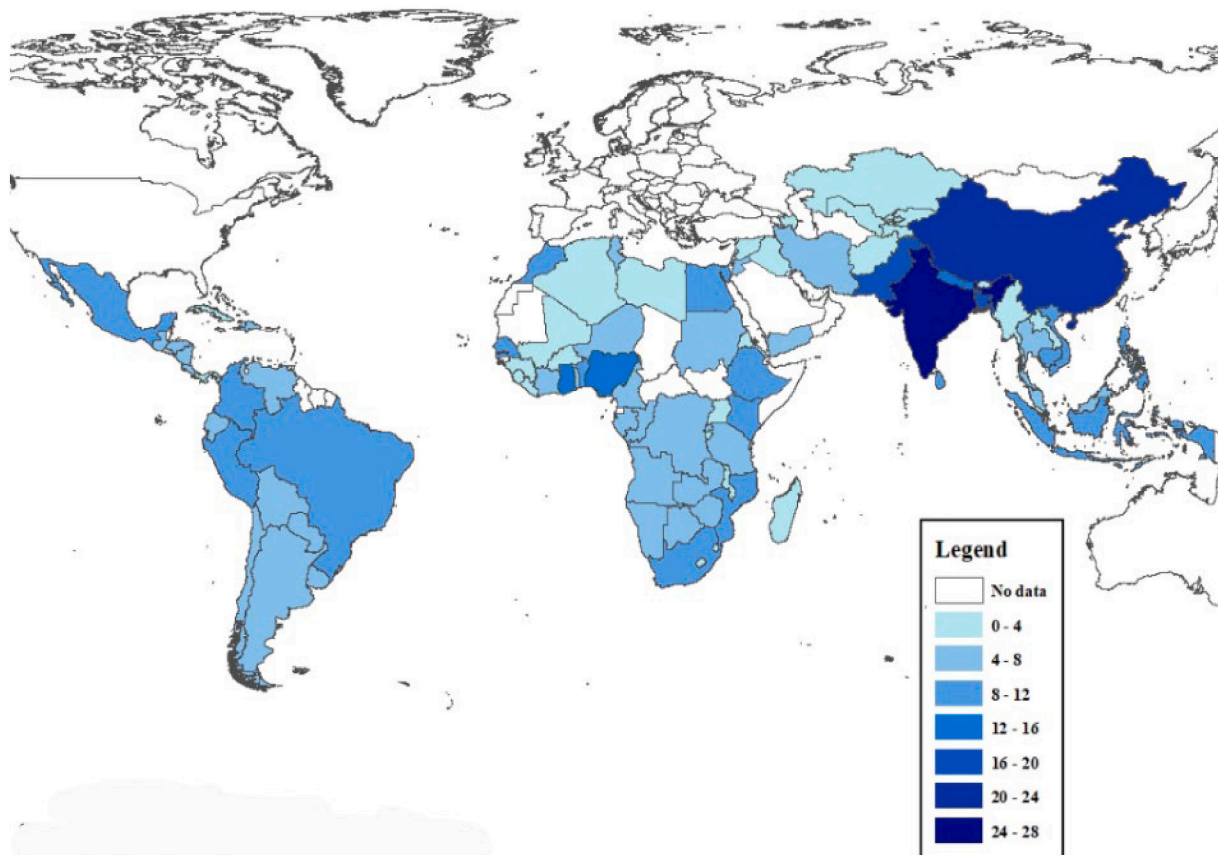


Fig. 5. Spatial distribution of publications on measuring energy poverty in developing countries.
Source: Done by authors.

and multidimensional forms [30]. It is built on a wide range of dimensions such as affordability, acceptability, reliability, quality, and adequacy, which interact with one another [31].

Fig. 6 presents the definition of the concept of energy access that includes various dimensions and comes from the demand and supply sides of energy services. On the demand side, the deprivation of access to energy services can be justified by three reasons [32]: (1) energy may not be available; (2) if energy is available, it may not be affordable; (3) and if energy is available and affordable, it may not be acceptable to the household. The concept of energy access refers to a combination of affordability, availability, and acceptability of modern energy sources needed to meet basic energy services.

In our knowledge of the concept of energy access, two issues have given rise to controversy [33,34]: the definition of basic energy services and identification of the minimum energy requirement for basic energy services. Energy services are considered as the services providing energy for household uses [35]. A broad range of energy services, including cooking and water heating, lighting, space cooling and heating, refrigeration, Information and Communication Technologies, education, entertainment, mechanical power, and transport, improve the standards of living and enhance social stability [5,36,37]. As mentioned by Pelz et al. [19], there is still some considerable controversy surrounding the definition of what are and are not to be regarded as basic energy services. For example, the concept of energy access by the German Society for International Cooperation (GIZ) considers three energy services such as cooking, lighting, and communication [34]. Nathan and Hari [32] conceptualize energy access as utilization of modern energy sources for cooking and lighting only.

The relevance of energy services depends on the energy context and climate and season [1,38,39]. While the energy context allows identifying energy services, which are and are not achieved in a specific country or geographical area, the climate and season bring to light the issue of heating and cooling services. Countries or regions differ in terms

of climate and season [39,40]. Accordingly, Culver [6] notes the difficulties in identifying a universal minimum energy requirement for basic energy services. The measures of the minimum energy requirement for basic energy services include useful energy consumption, end-use energy consumption and primary energy consumption.

On the supply side, energy access is physical access to the supply of energy services provided by Energy Services Companies [28,41]. The performances of energy services can be different depending on the scope, depth and method of finance [42]. The scope identifies the number of useful energy and/or final energy services. The depth takes into account the number of organizational activities providing the service. Finally, the method of finance brings to light internal financing, lease financing, third party financing, and project financing [43]. Drawing on this, the aspects of energy access include the reliability, quality, affordability, and adequacy of the energy services delivered to households [26].

3.1.2. Definition of energy poverty

Many international organizations and individual researchers have attempted to define the concept of energy poverty in developing countries. At this time of writing, there is no accepted definition. The United Nations Development Programme (UNDP) defines energy poverty as an absence of sufficient choice in accessing adequate, affordable, reliable, high quality, and environmental and health-friendly energy services to support economic development opportunities for the communities [44]. According to the IEA [45], energy poverty is a lack of access to electricity and dependence on the traditional use of biomass for cooking and heating. The Asian Development Bank defines it as an inability to cook with modern cooking fuels and the lack of minimum energy to meet basic needs and income-generating activities at sunset [4].

The researchers have also put forward various definitions of energy poverty in developing countries. Bazilian et al. [46] define the concept of energy poverty as a lack of access to affordable and high-quality

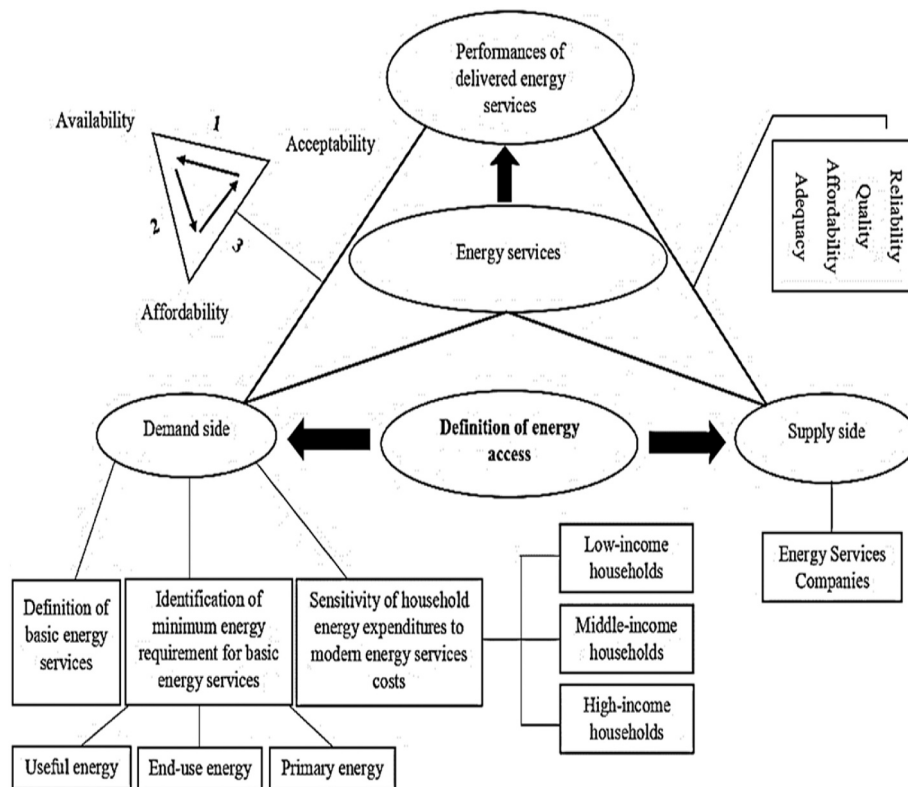


Fig. 6. Concept of energy access.
Source: Done by authors.

energy services. Other researchers, including Sadath and Acharya [1], Nussbaumer et al. [5], Day et al. [7], and Qurat-ul-Ann and Mirza [47], conceptualize energy poverty on the basis of Amartya Sen's capabilities approach as a theoretical framework [1,7]. The capability is a set of "doings and beings" that assesses the quality of life and represents a person's freedom to achieve various functioning combinations [48]. Energy poverty is considered as a lack of capabilities or absence of freedom to meet basic energy needs [49,50]. One can argue that this conceptualization of energy poverty gives a vision that tends to be absolutist. The absolutist notion of energy poverty refers, for example, to the need to use modern fuels and efficient energy technologies to perform household tasks. The household needs are, however, relative to society. The use of modern fuels and efficient energy technologies is heavily dependent on the status of people in society (e.g., social class and educational level). As an illustration, cooking or lighting with modern energy sources is dependent on the availability, reliability, and affordability of modern energy services and household preferences in terms of energy use or social and cultural factors [51]. Accordingly, the concept of energy poverty in developing countries takes an absolute form in the space of capabilities and is a relative notion in the space of commodities.

3.1.3. Energy poverty versus fuel poverty

There is a growing confusion in the literature between the concepts of energy poverty and fuel poverty [8,24,52,53]. Some researchers [54,55] use the concept of fuel poverty to describe the situation of energy poverty whereas others [38,56] consider fuel poverty as a measurement of energy poverty in developing countries [8]. Despite the similarities, many researchers in the field of energy poverty [8,52] highlight the differences between energy poverty and fuel poverty.

3.1.3.1. Definition of fuel poverty. Fuel poverty refers to the inability to afford enough modern energy services to live comfortable life [15]. In other words, people experiencing fuel poverty spends a much greater share of their incomes to afford adequate warmth [57]. The concept of fuel poverty is mostly applied in cold climate economies [15]. It is worth noting that the vast amount of literature on fuel poverty is from England, Ireland, and Scotland [8].

There is no single definition of fuel poverty. The United Kingdom was the first to establish a definition of fuel poverty in terms of the proportion of income spent on modern fuels to afford adequate warmth (i.e., 21 °C in the living rooms and 18 °C in the secondary rooms) in 2001 [58]. The inadequacy of this definition to capture the fuel poverty outcomes led the government of the United Kingdom to adopt a new definition of fuel poverty in 2013 called the "Low Income High Cost" approach [59]. The concept of fuel poverty is controversial in other countries like New Zealand, because of the absence of a precise definition adopted by the government [60]. In France, the concept of fuel poverty is a situation where a household suffers from insufficient energy sources due to the lack of resources or housing conditions [61]. Moore [62] attempts to seek a consensus on the definition of fuel poverty at the European level. As it turns out, the accommodating home standards differ from one country to another.

3.1.3.2. Conceptual difference. The concept of energy poverty concerns the problem of energy consumption in developing countries [8]. Developing countries face a lack of access to modern energy services [14]. All households from different income groups might use traditional as well as modern energy sources to meet their needs. Energy poverty in developing countries is a condition that affects all income groups [56,63]. A series of other factors, including availability and reliability of energy sources [64,65], household preferences, and social and cultural beliefs [51] can justify the lack of access to clean energy. There is a difference of opinion on the main causes of energy poverty. Some researchers [66,67] consider the price as the most important factor to

access modern energy sources whereas others [68,69] identify the availability and reliability of modern fuels as the main causes of energy poverty. According to Primc et al. [24], the concept of energy poverty is strongly associated with the concepts of energy markets, energy justice, energy use, energy policy, and social inequality. In contrast, a fuel poor household has already access to modern energy services, but it lacks sufficient income to achieve the minimum temperature thresholds [8]. The concept of fuel poverty relates to the concepts like energy efficiency and household socioeconomic characteristics [24] and focuses on the energy expenditures and inefficient buildings and appliances in the cold climate economies [8].

3.1.3.3. Conceptual similarity. Both concepts of energy poverty and fuel poverty have some similarities in terms of research sector (i.e., residential sector), group feature (i.e., low income), and main hazards (i.e., poverty, health, equity, and social development) [8]. Energy and fuel poor households have limited incomes, which would enable them to afford modern energy services. The lack of sufficient income leads to the use of inefficient energy fuels or technologies to meet basic needs (i.e., energy-poor) or suffering from the cold (i.e., fuel poor). The use of inefficient energy fuels or technologies [70,71] and exposure to the cold [62,72] cause respiratory health problems in the residential sector. Furthermore, the lack of access to modern energy services [73,74] or sufficient income to afford adequate warmth in the home [75,76] cause poverty and social exclusion.

3.2. Measurements of energy poverty in developing countries

Fig. 7 presents the energy metrics developed by international organizations and individual researchers. Various indicators are proposed to monitor and report progress towards achieving sustainable development goals: single indicators, dashboard indicators, and composites indices. The single indicators take into account the unidimensional aspect of energy poverty. These unidimensional measurements are divided into two approaches [38]:

- i) The engineering approach estimates the minimum energy requirement for basic energy services demanded by the household;
- ii) The economic approach focuses on the identification of the energy poverty line based on household income and expenditure.

It is important to emphasize that the single indicators are more suitable for the context of developed countries [11]. The most appropriate data were available for the developed countries than developing countries. The dashboard indicators are a set of non-aggregated indicators giving a broad picture of a country's energy system and measuring the progress in achieving the sustainable development goals [5,9]. The composite indices are a compilation of individual energy indicators into a single index on the basis of a theoretical framework in which variables must be rigorously selected, combined and weighted [77]. Also referred to as multidimensional indicators, composite indices seek to address the limitations of the single and dashboard indicators [5,9]. The multidimensional indicators are composed of two approaches:

- i) The household-based approach uses national survey data and covers the binary and multifaceted aspects of energy poverty. It demonstrates an analysis of the household deprivation in terms of access to modern energy services;
- ii) The target-based approach involves secondary data collection from various countries and evaluates the progress towards achieving the sustainable development goal relating to the global energy sector.

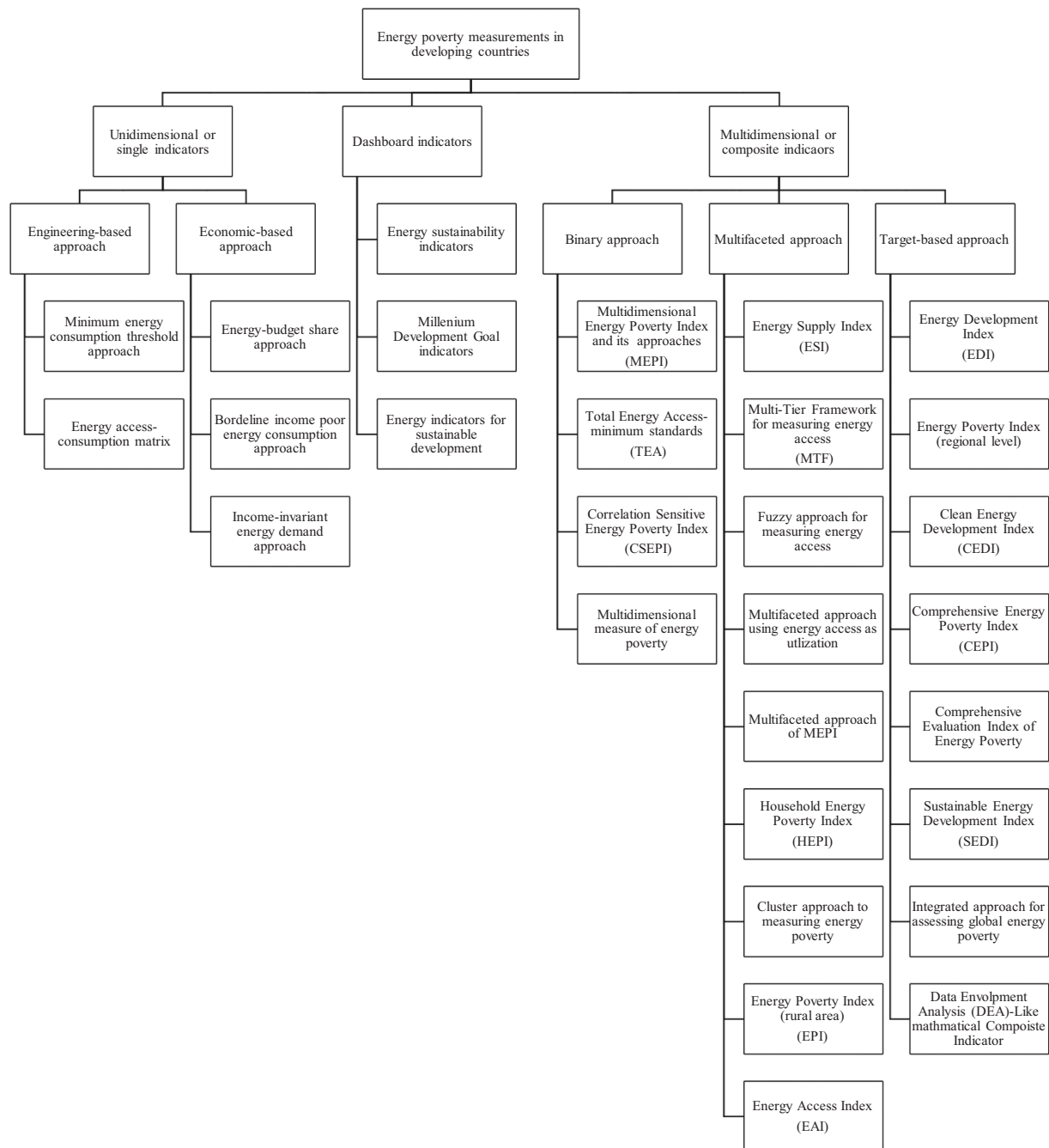


Fig. 7. Measurements of energy poverty in developing countries.
Source: Done by authors.

3.2.1. Single indicators

3.2.1.1. Engineering approach. The engineering approach for measuring energy poverty is embedded in the bottom-up approach that estimates the minimum energy requirement demanded by households. The energy requirement threshold is identified by considering the specifications of different energy sources, energy efficiency, and household size [32]. The early studies using the engineering approach include Parikh [78], Bravo et al. [79], Goldemberg [80], and Krugmann and Goldemberg [81] that focus on the final energy consumption. In contrast, Modi et al. [36] use useful energy to identify the minimum energy consumption threshold for the entire world (Table 2).

One significant limitation of the above-mentioned studies is that they only measure consumption and ignore the information on access to

different energy carriers [38]. The energy carriers differ in terms of quality, convenience, and utility. Pachauri and Spreng [38] construct an engineering approach, called the two-dimensional measure of energy poverty. It captures the information on access to different energy carriers and energy use. It focuses on three levels of access (i.e., biomass, electricity, and LPG) and four segments of energy use (i.e., bottom, lower, upper, and top). The two-dimensional measure of energy poverty is given in the form of an energy access-consumption matrix and divides the households into twelve groups.

Some limitations of the engineering approach can be offered for consideration. Setting the minimum energy required to satisfy basic needs is not an easy exercise, owing to the differences in terms of season and climate, cooking practices, and heating requirements between countries or regions [6,39]. Notwithstanding the difference in living

Table 2
Minimum energy requirement for energy services demanded by households.

Energy service	Minimum energy consumption threshold	Countries or regions	Reference
Cooking and lighting, producing foods (excluding human and animal power), and transport of foods and fuel	From 0.3 to 0.4 tce per capita	Developing countries	Parikh [78]
Lighting, cooking, space heating, space cooling, food preservation, water pumping, hot water, ironing clothes, recreation and social communication	27.4 kgOE per capita/month 32.1 kgOE per capita/month 200 kgOE per capita/year	Latin America Bangladesh Hot urban areas	Bravo et al. [79] Goldemberg [80] Krugmann and Goldemberg [81]
Services supplied by electricity including commercial and non-commercial energy use	1 kilowatt (kW) per capita	Developing countries	Goldemberg et al. [83]
Cooking and lighting	Much lower at 50 kgOE per capita/year	Universally	Modi et al. [36]
Cooking, lighting, space comfort, food preservation, and entertainment	100 W per capita	Rural areas	Reddy [84]

Source: Done by authors.

standards, the researchers try to calculate the minimum energy consumption threshold for some regions and the entire world. The engineering approach requires complex and interconnected data on energy services [82]. In addition, the minimum energy consumption threshold does not give a broader view of how energy poor are the population.

3.2.1.2. Economic approach. The economic approach involves setting an energy poverty line based on household income. The energy poverty line is obtained from the share of the household income devoted to energy consumption and the conventional poverty line. Three single indicators under the economic approach for measuring energy poverty are suggested in the literature: energy budget-share, borderline income poor energy consumption approach, and income-invariant energy demand approach.

Initiated by Leach [54], the energy budget-share approach is a single indicator that focuses on the percentage of income spent on energy consumption. It seeks to identify a sufficient level of income that households should spend on modern energy sources to live a decent life [85]. Foster et al. [55] develop the borderline income poor energy consumption approach to compute the average energy consumption according to the income poverty line established by the World Bank and UNDP. The energy budget-share approach as well as the borderline income poor energy consumption approach fails to capture the changes in household income.

In their innovative paper of 2011, Barnes et al. [56] address the drawbacks of both the energy-budget share approach and borderline income poor energy consumption approach. They suggest the income-invariant energy demand approach that assesses energy poverty using the energy poverty line defined at the threshold point in which energy consumption increases with household income (deciles). A household is considered energy-poor and income poor, at the same time, when its level of energy consumption (income) corresponds to that of its income (energy consumption). It may be considered non-poor in terms of income and still be poor in terms of energy due to the unavailability of modern energy services. Conversely, a household may be regarded as energy non-poor but its level of income still is below the income poverty line. An increase in its income must be above the poverty line meaning that the household has more choices for its energy expenditures. The

income-invariant energy demand approach gives a broader view of the energy-poverty nexus.

However, the income-invariant energy demand approach requires detailed price data across energy sources. The scarcity of data in developing countries does not facilitate the wide application of this approach [86]. To address the issue of the scarcity of data, Ye and Koch [86] follow an alternative approach that is conceptually similar to the income-invariant energy demand approach. The energy poverty line is derived from the baseline energy consumption and energy equivalence scale. The computation method of baseline energy consumption involves some assumptions concerning the reasonable living standards in a developing country and the specification of household size and composition. A semi-parametric model over household energy expenditure shares is used to estimate the energy equivalence scale. This alternative approach captures the Foster-Greer-Thorbecke poverty measures.

Although the economic approach is interesting, it focuses only on the economic aspect of energy poverty. Several reasons other than income or wealth can prevent households in developing countries from using modern energy sources, which are needed to provide modern energy services. These are the community factors, performances of the energy supply, and preference or traditional beliefs of households regarding modern energy sources [56,63]. It better suits the assessment of energy poverty in developed countries. Only a very limited number of studies have explored the suitability of the existing unidimensional measurements. Principally, this refers to the research studies conducted by Phoumin and Kimura [87] and Khundi-Mkomba et al. [88] who suggest a modified energy-budget share approach that is more suitable for developing countries. It captures both the affordability and availability of energy services. Furthermore, the economic approach fails to capture other facets of sustainable development, including environmental, social, technical, and institutional aspects. Accordingly, it cannot adequately assess the issues of energy poverty in developing countries [5,9].

3.2.2. Dashboard indicators

Many attempts have been made to assess energy poverty, based on economic, environmental, social, technical, and institutional sustainability. In the literature, three sets of indicators are developed to track and report the progress on energy issues at the national and global level: the energy sustainability indicators, millennium development goals indicators, and energy indicators for sustainable development [5,9].

The Latin America Energy Organisation, United Nations Economic Commission for Latin America and the Caribbean, and GIZ [89] carry out the first investigation into dashboard indicators. These international organizations suggest a set of energy sustainability indicators that includes the measurements of the targeted objectives for both residential and non-residential sectors. Three out of eight indicators, including electricity access rate, consumption of useful residential energy, and indoor air pollution treat the energy issues in the residential sector. Their objectives are to evaluate progress towards meeting basic energy needs, sufficient supply of modern energy services, higher air quality, and sustainable management of fuelwood. The remaining indicators need to examine the energy system with a particular accent on the macroeconomic, political and environmental issues. As regards the millennium development goals indicators, sixty indicators divided into eight goals assess progress related to eradicating extreme poverty and hunger, achieving universal primary education, promoting gender equality and empowering women, reducing child mortality, improving maternal health, combating diseases, ensuring environmental sustainability, and developing a global partnership for development [90]. The millennium development goals indicators do not explicitly analyze the energy issues, but the targeted goals that they are associated with cannot be achieved without sufficient and affordable energy [4]. The IAEA in partnership with the IEA, United Nations Department of Economics and Social Affairs, Eurostat, and European Environmental Agency propose

the set of energy indicators for sustainable development. The energy indicators for sustainable development are designed to assess the current conditions of the country's energy system and identify suitable energy policy options towards achieving sustainable development [91]. The set of energy indicators for sustainable development covers thirty indicators, including sixteen, four, and ten indicators of the economic, social, and environmental dimensions of sustainable development respectively.

The dashboard indicators are a set of non-aggregated indicators, which involve secondary data collection and require a thorough analysis of the issue of energy poverty in all sectors. As highlighted by many researchers [5,9], the dashboard indicators do not facilitate benchmarking between countries or regions.

3.2.3. Multidimensional measurements

3.2.3.1. Binary approach. The binary approach of energy poverty is an access-based approach, which identifies households with or without access to modern energy sources [32]. The binary approach is composed of several composite indices that assess both demand and supply side of energy services: the Multidimensional Energy Poverty Index (MEPI) and its approaches, Multidimensional measure of energy poverty, Total Energy Access (TEA) approach, and Correlation Sensitive Energy Poverty Index (CSEPI).

The MEPI is the most commonly used indicator for the assessment of energy poverty in developing countries [9]. This composite index is developed by Nussbaumer et al. [5] using the structure of the Multidimensional Poverty Index (MPI) suggested by the Oxford Poverty Human Initiatives. The MEPI focuses on the deprivation of access to modern energy services such as cooking, lighting, cooling, entertainment/education, and communication. It does not take into account the dimensions of energy poverty such as mechanical power, climate factors, and mobility. Nussbaumer et al. [5] recognize the importance of the mechanical power, but it turns out that there is a lack of reliable data. Concerning the way to deal with other limitations, a wealth of literature has started to emerge since 2019 investigating various approaches of MEPI. Mendoza et al. [92] use the indicators of space cooling to capture the climate factors. Moreover, they define another indicator of communication such as ownership of the computer, which is not taken into consideration within the Nussbaumer et al.'s multidimensional approach. Robles-Bonilla et al. [93] and Cedano et al. [94] include the thermal comfort in order to deal with the limitations of MEPI regarding climate factors. The thermal comfort is calculated based on the bioclimatic regionalization, which addresses the geographical and weather conditions in a developing country. Qurat-ul-Ann and Mirza [47] integrate relevant dimensions of energy poverty in their approach, including space heating and temperature regulation and mobility. Unlike these proposed approaches of MEPI, Ssenono et al. [95] construct a MEPI that assesses both demand and supply sides of energy poverty. More specifically, they incorporate various aspects of energy poverty such as affordability, availability, capacity, quality, expose, safety, convenience, and deprivation of household appliances.

The MEPI and aforementioned approaches are designed to assess energy poverty at the national level, but fail to properly inform policymakers about the issues of energy poverty in rural areas. Rural households use different combinations of modern and traditional energy sources called energy mixes [96]. The specific insights into rural energy access provided by these indices and aimed at influencing policies may be misleading. This enhances the need to develop an area-specific approach to dealing with the gap between policies and actions [34]. Little research looks at the assessment of energy poverty in rural areas. This refers primarily to the research study of Gafa and Egbendewe [97]. Both authors propose another approach of MEPI that is suitable for rural areas.

In the framework of assessing energy poverty using the MEPI, the

issue that arises is how to assess the relative importance of the dimensions of energy poverty. Some studies use the subjective method of assigning weights to the dimensions of energy poverty that is suggested by Nussbaumer et al. [5,98,99]. Others provide an adjusted method of weighting [100,101]. The subjective methods are arbitrary methods that reflect a value judgment per se on the relative importance of the indicators of energy poverty, thereby generating imprecision and uncertainty in assessing energy poverty [16,32]. For this reason, Sadath and Acharya [1] suggest a modified MEPI using a methodology that combines unequal and equal weighting for the sensitivity analysis. Zhang et al. [10] adopt this approach to construct a multidimensional measure of energy poverty, which capture the affordability and accessibility of modern energy services. Following the approach of Mendoza et al. [92], Jayasinghe et al. [102] use both subjective (i.e., equal weights and adjusted weights suggested by Mendoza et al.) and scientific (i.e., PCA) methods to assign the weights to the dimensions of energy poverty and test the sensitivity of the MEPI. Moreover, Qurat-ul-Ann and Mirza [103] estimate the weights of the dimensions of energy poverty based on the Multi-Criteria Analysis (MCA) technique. This approach involves several steps, including screening, prioritizing, selecting, and ranking alternatives weights based on multiple generally conflicting criteria. The relative importance of each criterion is determined by applying the judgment of experts.

In its seminal report on poor people's energy outlook of 2010, Practical Action proposed the TEA approach and then refined it in 2012 using the structure of the MEPI. In a similar vein, Pablo et al. [104] develop another multidimensional approach. The TEA approach relies on the minimum energy requirement for a household to meet basic energy services whereas the MEPI approach of Pablo et al. [104] focuses only on the economic aspects of energy poverty. The MEPI and TEA approaches are based on an arbitrary threshold to identify energy-poor people. Nussbaumer et al. [5] set the energy poverty line at 0.33, but it may vary within the range of [0.2, 0.4]. Rafi et al. [11] consider only 0.3 and 0.7 as alternative energy poverty lines. Another issue for consideration is that these binary approaches may be sensitive to the choice of the energy poverty line. Drawing from the work of Rippin [105] on the MPI, Bensch [106] develop a modified MEPI called CSEPI to avoid the use of arbitrary thresholds for identifying energy-poor.

Recent publications reveal the limitations of the binary approach [30,32,107]. The concept of energy poverty has not a binary nature that identifies households with or without access to modern energy sources [32]. The binary approach fails to take into account the relative efficiency of energy sources and technologies in an explicit manner. The IEA [45] provide the classification of energy sources and technologies for cooking and lighting according to their efficiency (Fig. 8). The use of electric stoves is, for example, more efficient in terms of quality, affordability, environmental and health-friendly of the delivered energy services than that of improved biomass cookstove to produce the same amount of cooking, although both energy technologies are considered as modern energy technologies. The binary approach does not offer broader insights into energy poverty in the residential sector [30]. Energy poverty is, rather, a concept that characterises itself in different degrees and shapes.

3.2.3.2. Multifaceted approach. The multifaceted approach assesses the different forms of energy poverty in the residential sector, whether at the dimensional level [30,37] or aggregated level [107,108]. The composite indices under the multifaceted approach are proposed to fill the gap with the binary approach by integrating the availability, acceptability, affordability, and safety of energy sources, and efficiency of energy technologies into the assessment of energy poverty.

The Practical Action [37] is the first to assess the multifaceted aspects of energy poverty in developing countries by developing the Energy Supply Index, which focuses on different levels (from 0 to 5) of access to clean energy fuels and technologies for cooking, lighting, and

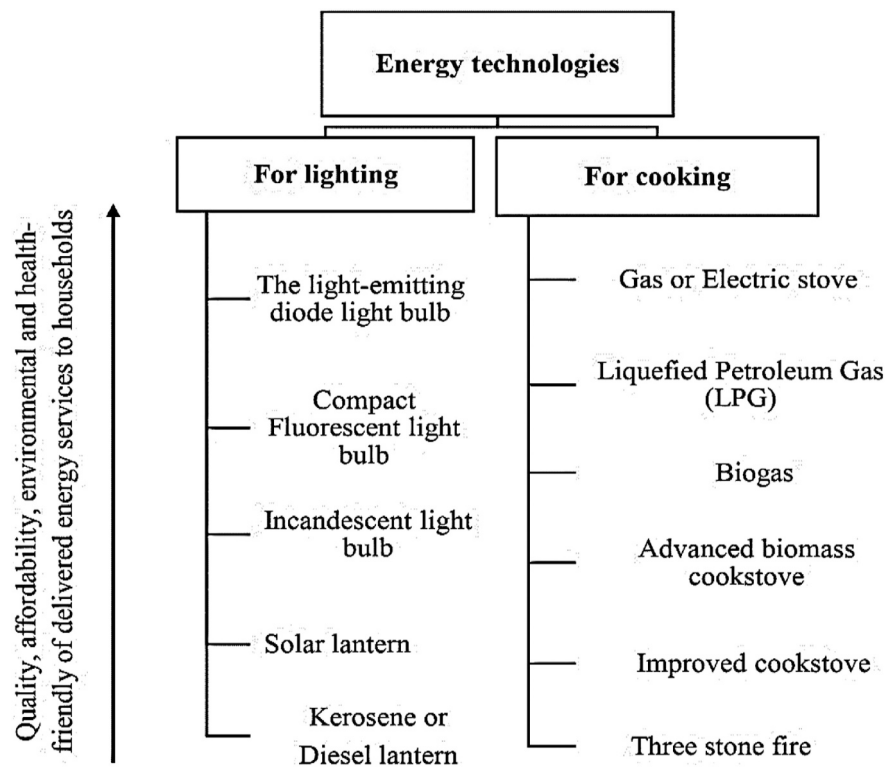


Fig. 8. Efficient energy sources and technologies for cooking and lighting.
Source: Adapted from IEA [45].

mechanical power. The World Bank, through the Energy Sector Management Assistance Programme, proposes a more generalized approach – the Multi-Tier Framework (MTF) – that captures not only the supply side of energy services but also the demand side to fill the research gap. The MTF includes different levels of access to electricity and clean cooking fuels or technologies called tier (from 0 to 5). The World Bank approach to measuring energy access provides more information about households that lead to better policy decisions: tracking progress in sustainable development goals, setting country-specific realistic targets for universal access, analyzing the relationship between energy access and other sectors and the current energy usage, and providing other relevant supply and demand data [30]. However, the operationalization of the MTF is complex at the global tracking level. The MTF does not take into account the national objectives and may fail to gain acceptance in national planning [34,92].

Other approaches, which are more aligned with the national strategy for eradicating energy poverty, have started to emerge in the literature. Robles-Bonilla et al. [108] use a methodology based upon the fuzzy classification. The fuzzy approach for measuring energy access includes gathering information on energy consumption in Mexico using linguistic variables (i.e., low, medium, and high-energy access). It focuses on expenditure consumption for electricity, cooking fuels, and transport. As a result, the fuzzy approach for measuring energy access cannot capture the availability and acceptability of modern energy as well as the environmental and health hazards related to the use of traditional fuels. Nathan and Hari [32] conceptualize energy access as a combination of the affordability, acceptability, and availability of modern energy sources. They suggest a methodology for assessing energy poverty in India using the conceptualization of energy access as utilization. Their proposed approach identifies four groups of households in terms of energy consumption for lighting and cooking: energy non-poor, transitional energy-poor, moderate energy-poor, and extreme energy-poor. The advantage of this multifaceted approach is that it avoids the use of weights and thresholds to compute the depth and severity of energy

poverty. Nevertheless, its scope is restricted to cooking and lighting, and thereby fails to capture the importance of the use of household appliances. The approach of Nathan and Hari [32] may not provide effective support for the governmental policies to address energy issues within a broader context of energy poverty. In the context of assessing energy poverty in Kenya, Olang et al. [109] suggest a multifaceted approach of MEPI that has a broad scope. They consider three scenarios for representing the multidimensional energy poverty in the residential sector, namely acute energy poverty, moderate energy poverty, and low energy poverty. This multifaceted approach may be subject to criticism for assigning subjective weights to the different scenarios.

The Household Energy Poverty Index (HEPI) is also designed to assess energy poverty in India. It addresses the drawbacks related to the restricted scope of Nathan and Hari's approach and fills the research gaps concerning the integration of the availability and affordability indicators into energy poverty assessment and the use of scientific methods (i.e., PCA) for assigning the weights to energy poverty indicators [107]. The HEPI relies on a set of indicators and captures the contribution of each indicator to the overall energy poverty and the contribution of each indicator in the household energy poverty. The clustering approach is used to avoid the arbitrary threshold for identifying the following groups of households: least, less, more, and most energy-poor. However, the HEPI does not show the kind of energy sources used by the different groups of households. In other words, it does not give a clear interpretation of least, less, more, and most energy-poor. García Ochoa and Ed [110] propose a cluster approach for measuring energy poverty in Mexico that provides a broader interpretation based on the privation of energy services.

The above-mentioned approaches describe and assess the multifaceted nature of energy poverty at the national level, but do not properly address the inequality of energy poverty between urban and rural areas [34]. These approaches do not reflect the real picture of energy access in rural areas. Few researchers address the problem of energy access in rural areas using multifaceted approach. Mirza and Szirmai [96] suggest

the EPI for rural Pakistan, which relies on the energy inconvenience (i.e., the frequency of buying or collecting, household involvement, and time spent on energy collection per week) and energy shortfall. Drawing on the literature of EPI for rural Pakistan and MEPI, Nayan Yadava and Sinha [34] develop the EAI to assess energy poverty in the forest fringe villages in India. The EAI focuses on the deprivation of access to modern energy services.

3.2.3.3. Target-based approach. Understanding the differences between the household-based approach and target-based approach is crucial when it comes to setting guidance on national energy planning that focuses on the residential sector and global energy sector. Energy poverty is more than just a lack of access to modern energy services in the residential sector [111]. The challenges of tackling energy poverty involve providing universal access to modern energy services, reducing global energy intensity, and increasing global renewable energy use to primary energy supply [4]. The target-based approach is accordingly designed to evaluate the progress towards achieving the sustainable development goals relating to the global energy sector. Various indicators are put forward to assess developments in the global energy sector: the Energy Development Index (EDI) and its approaches, Sustainable Energy Development Index (SEDI), Comprehensive Energy Poverty Index (CEPI) and its approaches, Energy Poverty Index (EPI) for assessing regional energy poverty, and integrated approach for assessing global energy poverty.

First proposed by the IEA [45] in 2004 and then refined in 2010, the EDI is aimed at explaining the role of energy in human development and tracking the energy transition to the use of modern fuels in developing countries. The EDI is based on four indicators, including per capita commercial energy consumption, per capita electricity consumption in the residential sector, the share of modern fuels in total residential sector energy use, and the share of the population with access to electricity. It leaves neutral judgment on the weights of energy poverty indicators [112]. Other approaches of EDI have started to emerge in the literature. These involve the research studies of Wang et al. [113] and Banerjee et al. [114]. On the one hand, Wang et al. [113] develop the Clean Energy Development Index (CEDI) to promote the crucial role of renewable energy towards achieving Sustainable Development Goal 7 (SDG7). On the other hand, Banerjee et al. [114] use the structure of UNDP's method of constructing the Human Development Index to suggest an adjusted EDI.

The aforementioned approaches do not reflect an essential understanding of the issues involved in the use of different energy sources and technologies such as affordability and sustainability issues [9,115]. Some attempts have been made to address these limitations. By focusing on the national objective with respect to the energy accessibility and affordability, Bonatz et al. [12] develop an EPI for assessing energy poverty at the regional level. Khanna et al. [115] propose the CEPI to capture the affordability, availability, and cleanliness of modern energy fuels and technologies. Finally, Iddrisu and Bhattacharyya [9] develop the SEDI to fill the gap related to sustainability issues (i.e., the technical, economic, social, environmental, and institutional sustainability). However, from a methodological perspective, the subjective weights are used to assign the weights to the components of EPI, CEPI, and SEDI. These composite indices may be subject to criticism since the subjective weights are not based on scientific criteria [1,16,32].

Considering the recent advances in assigning the weights to indicators of energy poverty, the scientific approaches are increasingly used as they help to avoid the imprecision and uncertainty related to the subjective weights [107,116]. Che et al. [116] use the multiplicative integrated method combining objective (obtained using rough set theory) and subjective weights to develop an integrated approach for assessing global energy poverty, which focuses on the components of CEPI. Li et al. [117] use the Data Envelopment Analysis (DEA) and entropy methods to measure another approach of CEPI based on four

indicators such as energy availability, energy access, energy consumption, and clean energy affordability. Wu et al. [118] also use the DEA model to develop a DEA-Like mathematical composite indicator. Others researchers such as Zhao et al. [119] consider the entropy method. Both authors follow the research study of Wang et al. [120] and propose a CEPI that is more aligned with the national energy system in China. Furthermore, Chien et al. [121] use the MCA and DEA as methods of weighting. They include seventeen indicators in their assessment of energy poverty at the macro-level, which are the same as that of the SEDI.

Despite the issues relating to the weights of the indicators of energy poverty, the SEDI relies on a diverse set of indicators that covers the dimensions of the CEPI and its approaches, DEA-Like mathematical composite indicator, and integrated approach for assessing global energy poverty as well as other aspects of the sustainable development, including technical and institutional sustainability.

4. Conclusion

The conclusions that can be drawn from this review are on four levels. First, developing countries experience different energy issues according to the climate and season, household preferences, accommodating home standards, and social and cultural beliefs. It would be difficult to provide a definition of energy poverty that could be sufficiently broad to encompass all these factors. One should accordingly consider the relativist rather than the absolutist notion of energy poverty.

Second, the concept of energy poverty should not be confused with the concept of fuel poverty, even though there are similarities between both concepts in terms of the research sector, group feature, and main hazards. The concept of energy poverty addresses the issues of availability, affordability, acceptability, reliability, quality, and adequacy of energy services whereas the concept of fuel poverty focuses on the lack of sufficient income to achieve the minimum temperature thresholds (e.g., 21 °C in the living rooms and 18 °C in the secondary rooms) in cold climate countries. In this context, the emphasis is put on energy consumption in the residential sector. The issue of energy poverty in developing countries is also related to energy use at the macro-level. The energy challenge encompasses the use of non-renewable energies that cause major environmental hazards. In this particular case, developing countries are facing a lack of appropriate action for promoting renewable energies, improving air-quality pollution, and mitigating climate change.

Third, several indicators, including single indicators, dashboard indicators, and composite indices are used for communicating energy poverty issues and tracking progress towards the achievement of SDG7. The composite indices are more generally applied in assessing energy poverty because of their capacity to aggregate all relevant information on basic energy services and facilitate benchmarking between countries. The review found that although the binary approaches are widely used, the recent research on the measurements of energy poverty in developing countries tends to be oriented towards analysis of the multifaceted aspects of energy poverty at the household level. Furthermore, few approaches that best suit the assessment of rural energy poverty are developed in the literature. In the framework of setting guidance on national energy planning that focuses on the global targets of eradicating energy poverty, the SEDI appears to be the most appropriate indicator for assessing energy poverty in developing countries. The SEDI relies on a diverse set of indicators that covers the dimensions of the other composite indices of the target-based approach as well as the sustainability dimensions such as technical and institutional sustainability. However, developing countries are much more concerned with the issue of income poverty than the phenomenon of energy poverty [103]. This contributes to the delays in the implementation of the target-based approach. Another issue is that the scarcity of data prevents the widespread use of the measurements of energy poverty in developing

countries.

Fourth, the proposed new approaches for assessing energy poverty in developing countries (e.g., EAI, HEPI, and CSEPI) deal with the imprecision and uncertainty related to subjective weights and arbitrary thresholds. The scientific approaches (e.g., DEA, MCA, and PCA) are used as the methods to assign the weights to indicators of energy poverty. Many studies conceptualize energy access (e.g., Nathan and Hari [32]) or use scientific methods (e.g., Bensch [106]; Gupta et al. [107]) to avoid the arbitrariness in the choice of the energy poverty line. As regards the identification of the energy poverty line for assessment of energy poverty at the unidimensional level, the use of income deciles provides a greater classification of extent of energy poverty [56] than the energy-budget share [54] and official income poverty line [55].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We are most grateful to the United Nations University World Institute for Development Economics Research (UNU-WIDER) for the opportunity to utilize resources and facilities for this research, [605UU-00000000000000000000000000002394, 2019]. We thank the anonymous reviewers whose comments/suggestions helped improve and clarify this manuscript. We are also thankful to Frank Adusah Poku (University of Ghana) for his valuable review of the previous version of this manuscript. Finally, we want to thank Amadou Ndongo Thiam, Moustapha Cisse Fall, Mamadou Ndiaye Sene and Jack K. Simukwai for their supports.

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